

Rationalising the Denominator

AQA · KS4 Year 10–11

Common Mistakes to Avoid:

- Must multiply BOTH top and bottom by the same thing.
- For $(a + b)$ use conjugate $(a - b)$, not just b .
- $(-3)^2 = 9$, not -3 .
- Don't forget to simplify at the end.

Method Steps:

1. Simple: multiply top and bottom by a
2. $2 - a \times a = a^2$ denominator rational
3. 3 Complex: multiply by the conjugate $(a - b)$
4. $4 (a + b)(a - b) = a^2 - b^2$
5. 5 Simplify the fraction

Worked Example:

Rationalise $5/(3 + 2)$.

Conjugate: $(3 - 2)$

Top: $5(3 - 2) = 15 - 5 = 10$

Bottom: $(3 + 2)(3 - 2) = 9 - 2 = 7$

Answer: $(15 - 5)/7$

Visual Aid: Decision tree: Is the denominator just a ? Yes multiply by a/a . No Is it $(a \pm b)$? Yes multiply by conjugate $(a \mp b)/(a \mp b)$.

Question 1 [2 marks]

Rationalise: a) $\frac{1}{3}$ b) $\frac{1}{5}$ c) $\frac{1}{7}$ d) $\frac{1}{2}$

Question 2 [2 marks]

Rationalise: a) $\frac{4}{2}$ b) $\frac{6}{3}$ c) $\frac{10}{5}$ d) $\frac{3}{6}$

Question 3 [3 marks]

Rationalise and simplify: a) $\frac{12}{6}$ b) $\frac{2}{8}$ c) $\frac{5}{20}$ d) $\frac{(2-3)}{12}$

Question 4 [3 marks]

Rationalise: a) $\frac{1}{(1+2)}$ b) $\frac{1}{(3-5)}$ c) $\frac{2}{(4+3)}$

Question 5 [3 marks]

Rationalise and simplify: a) $\frac{6}{(3+1)}$ b) $\frac{10}{(5-3)}$ c) $\frac{(1+2)}{(1-2)}$

Question 6 [4 marks]

Rationalise: a) $\frac{(3+5)}{(2-5)}$ b) $\frac{(7+2)}{(7-2)}$

Question 7 [4 marks]

a) Show $\frac{1}{(5-2)} - \frac{1}{(5+2)} = 4$. b) Express $\frac{(2+3)}{(2-3)}$ as $a+b\sqrt{3}$.

Question 8 [5 marks]

a) Rationalise $(\sqrt{3} + \sqrt{2})/(\sqrt{3} - \sqrt{2})$ fully. b) Rectangle area 6 cm^2 , width $(\sqrt{3} - 1)$. Exact length?
